

SEQUENCES, CONTAINERS, OBJECTS, & LINKED LISTS

COMPUTER SCIENCE MENTORS 61A

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1 Sequences

Sequence Overview

Sequences are ordered data structures that have length and support element selection. Here are some common types of sequences you'll be dealing with in this class:

- Lists: `[1, [2], 'a', lambda x: 5]`
- Tuples: `(1, (2,), 'a', lambda x: 5)`
- Strings: `'Hello World!'`

While each type of sequence is different, they all share a common interface for manipulating and accessing their data:

- **Item selection:** Use square brackets to select an element at an index:

```
(3, 1, 2)[0] # returns 3
"Hello"[-1]  # returns "o"
```

- **Length:** The built-in `len` function returns the length of a sequence:

```
len((1, 2)) # returns 2
```

- **Concatenation:** Sequences can be concatenated with the `+` operator, which returns a *new* sequence:

```
[1, 2] + [3, 4] # returns [1, 2, 3, 4]
```

- **Membership:** The `in` operator tests for sequence membership:

```
1 in (1, 2, 3)          # returns True
5 not in (1, 2, 3)       # returns True
"apple" in "snapple"    # returns True
```

Membership in Strings vs. Lists and Tuples: As a short aside, while the `in` operator works the same for lists and tuples, checking if an element is contained within the list/tuple container, the `in` operator instead for strings checks for direct substrings rather than the existence of distinct elements within the string.

- **Looping:** Sequences can be looped through with `for` loops:

```
>>> for x in [1, 2, 3]:
...     print(x)
1
2
3
```

- **Aggregation:** Common built-in functions—including **sum**, **min**, and **max**—can take sequences and aggregate them into a single value:

```
max((3, 4, 5)) # returns 5
```

- **Slicing:** Slicing is a way to create a copy of all or part of a sequence. The general syntax for slicing a sequence `seq` is as follows:

```
seq[<start index>:<end index>:<step size>]
```

This evaluates to a new sequence that includes every element starting at `<start index>` and up to and *excluding* `<end index>` in `seq`, taking steps of size `<step size>`.

If we do not supply `<start index>` or `<end index>`, it will start at the beginning of the sequence and include every element up to and including the end of the sequence.

```
>>> lst = [1, 2, 3, 4, 5]
>>> lst[2:]
[3, 4, 5]
>>> lst[:3]
[1, 2, 3]
>>> lst[::-1]
[5, 4, 3, 2, 1]
>>> lst[1:2]
[2, 4]
```

List comprehensions create a new list by processing an iterable, such as a list, tuple, string, or range. The syntax for a list comprehension is

```
[<expression> for <element> in <sequence> if <condition>]
```

We could equivalently write the following:

```
lst = []
for <element> in <sequence>:
    if <condition>:
        lst = lst + [<expression>]
```

The **if** `<condition>` filter statement is optional. The following list comprehension doubles each odd element of `[1, 2, 3, 4]`:

```
>>> [i * 2 for i in [1, 2, 3, 4] if i % 2 != 0]
[2, 6]
```

Equivalent in **for** loop syntax:

```
lst = []
for i in [1, 2, 3, 4]:
    if i % 2 != 0:
        lst = lst + [i * 2]
```

Sequence Practice

1. What would Python display? Draw box-and-pointer diagrams for the following:

```
>>> a = [1, 2, 3]
>>> a
```

```
>>> a[2]
```

```
>>> a[-1]
```

```
>>> b = a
>>> a = a + [4, [5, 6]]
>>> a
```

```
>>> b
```

```
>>> c = a
>>> a = [4, 5]
>>> a
```

```
>>> c
```

```
>>> d = c[3:5]
>>> c[3] = 9
>>> d
```

```
>>> c[4][0] = 7
>>> d
```

```
>>> c[4] = 10
>>> d
```

```
>>> c
```

2. **Tuples.** What would Python display? Run these statements in order. If a statement causes an error, identify the error and continue with the next statement. For `(3,)` and `(3)`, also name the type.

Creating tuples

```
>>> t: tuple[int, str, float, bool, None] = (  
...     3, "csm", 2.5, True, None  
... )  
>>> t
```

```
>>> 4, 5, 6, 7
```

```
>>> (3,)
```

```
>>> (3)
```

```
>>> ()  
>>> len(())
```

Slicing and converting

```
>>> t[1:4]
```

```
>>> t[::-1]
```

```
>>> list(t)
```

```
>>> list("csm")  
>>> list(())
```

Combining tuples

```
>>> (4, 5) + (6, 7)
```

```
>>> (3,) * 3
```

```
>>> (3) * 3
```

```
>>> t[0] = 4
```

Dictionary keys

```
>>> locations = {(4, 5): "CSM"}
```

```
>>> locations[(4, 5)]
```

```
>>> locations[[4, 5]] = "office hours"
```

```
>>> locations[(4, [5])] = "office hours"
```

3. Write a function `duplicate_list`, which takes in a list of positive integers and returns a new list with each element `x` in the original list duplicated `x` times.

```
def duplicate_list(lst: list[int]) -> list[int]:
```

```
    """
```

```
    >>> duplicate_list([1, 2, 3])
```

```
    [1, 2, 2, 3, 3, 3]
```

```
    >>> duplicate_list([5])
```

```
    [5, 5, 5, 5, 5]
```

```
    """
```

```
    _____
```

```
    for _____:
```

```
        for _____:
```

```
            _____
```

```
    _____
```

4. Write a function that takes in a list `nums` and returns a new list with only the primes from `nums`. Assume that `is_prime(n)` is defined. You may use a **while** loop, a **for** loop, or a list comprehension.

```
def all_primes(nums):
```

5. Fill in the methods below according to the doctests.

```
def gen_list(n):
    """
    Returns a nested list structure of n elements where the
    ith element is a list from 0 (inclusive) to i (exclusive).
    >>> gen_list(3)
    [[0], [0, 1], [0, 1, 2]]
    >>> gen_list(5)
    [[0], [0, 1], [0, 1, 2], [0, 1, 2, 3], [0, 1, 2, 3, 4]]
    """
    return _____
```

For an additional challenge, try out the following:

```
def gen_increasing(n):
    """
    Returns a nested list structure of n elements where the
    ith element of each list is one more than the previous
    element (even if the previous is in a prior sublist).
    >>> gen_increasing(3)
    [[0], [1, 2], [3, 4, 5]]
    >>> gen_increasing(5)
    [[0], [1, 2], [3, 4, 5], [6, 7, 8, 9], [10, 11, 12, 13,
    14]]
    """
    return _____
```

Hint: You can sum ranges. E.g. `sum(range(3))` gives us $0 + 1 + 2 = 3$.

2 Containers and Dictionaries

6. Write a function `count_t`, which takes in a dictionary `d` and a string `word`. The function should count the instances of the letter "t" in `word` and add a key-value pair to the dictionary. The key will be `word` and the value will be the number of "t"s in `word`

```
def count_t(d: dict[str, int], word: str) -> None:
    """
    >>> words = {}
    >>> count_t(words, "tatter")
    >>> words["tatter"]
    3
    >>> count_t(words, "tree")
    >>> words
    {'tatter': 3, 'tree': 1}
    """
```

```
    for _____:
        if _____:
```

7. Complete the function `snapshot`, which takes a single-argument function `f` and a list `snap_inputs` and returns a "snapshot" of `f` on `snap_inputs`. A "snapshot" is a dictionary where the keys are the provided `snap_inputs` and the values are the corresponding outputs of `f` on each input.

```
def snapshot(f, snap_inputs):
    """
    >>> snapshot(lambda x: x**2, [1, 2, 3])
    {1: 1, 2: 4, 3: 9}
    """
```

```
    snap = _____
    _____:
        _____

    return snap
```

3 Objects and Data Abstraction

8. (a) What is an **Abstract Data Type**?

(b) What is the relationship between a class and an ADT?

9. What would Python display?

```
class Foo(object):
    x = 'bam'

    def __init__(self, x):
        self.x = x

    def baz(self):
        return type(self).x + self.x

class Bar(Foo):
    x = 'boom'

    def __init__(self, x):
        Foo.__init__(self, 'er' + x)

foo = Foo('boo')
```

(a) `>>> Foo.x`

(b) `>>> foo.x`

(c) `>>> foo.baz()`

(d) `>>> Foo.baz()`

(e) `>>> Foo.baz(foo)`

(f) `>>> bar = Bar('ang')`
`>>> Bar.x`

(g) `>>> bar.x`

(h) `>>> bar.baz()`

For each of the following problems, assume linked lists are defined as follows:

```
class Link:
    empty = ()
    def __init__(self, first, rest=empty):
        assert rest is Link.empty or isinstance(rest, Link)
        self.first = first
        self.rest = rest

    def __repr__(self):
        if self.rest is not Link.empty:
            rest_repr = ', ' + repr(self.rest)
        else:
            rest_repr = ''
        return 'Link(' + repr(self.first) + rest_repr + ')'

    def __str__(self):
        string = '<'
        while self.rest is not Link.empty:
            string += str(self.first) + ' '
            self = self.rest
        return string + str(self.first) + '>'
```

10. What will Python output? Draw box-and-pointer diagrams along the way.

```
>>> a = Link(1, Link(2, Link(3)))
```

```
>>> a.first
```

```
>>> a.first = 5
```

```
>>> a.first
```

```
>>> a.rest.first
```

```
>>> a.rest.rest.rest.rest.first
```

```
>>> a.rest.rest.rest = a

>>> a.rest.rest.rest.rest.first

>>> repr(Link(1, Link(2, Link(3, Link.empty))))

>>> Link(1, Link(2, Link(3, Link.empty)))

>>> str(Link(1, Link(2, Link(3))))

>>> print(Link(Link(1), Link(2, Link(3))))
```

11. Write a function `linkify` that, given a list `lst`, returns a linked list with the same values. You may assume that there will be no nested lists.

Try writing this function both recursively and iteratively.

```
def linkify_rec(lst):  
    """  
    >>> lst = [0, 1, 2, 3]  
    >>> linkify_rec(lst)  
    Link(0, Link(1, Link(2, Link(3))))  
    """  
    if _____:  
        _____  
  
    else:  
        _____  
  
def linkify_iter(lst):  
    _____  
  
    for _____:  
        _____  
    _____
```

12. Write a function `combine_two`, which takes in a linked list of integers `lnk` and a two-argument function `fn`. It returns a new linked list where every two elements of `lnk` have been combined using `fn`.

```
def combine_two(lnk, fn):  
    """  
    >>> lnk1 = Link(1, Link(2, Link(3, Link(4))))  
    >>> combine_two(lnk1, add)  
    Link(3, Link(7))  
    >>> lnk2 = Link(2, Link(4, Link(6)))  
    >>> combine_two(lnk2, mul)  
    Link(8, Link(6))  
    """  
    if _____:  
        return _____  
    elif _____:  
        return _____  
    combined = _____  
    return _____
```

5 Optional Extension: Tree-Recursive Counting

13. Gabe's Donut shop has an unlimited supply of f different flavors of donuts. Adit wants to buy a box containing d donuts. Complete the skeleton for the function `donut`, which determines the number of possible ways there are for Adit to select his d donuts from the f flavors. You may assume that d and f are non-negative integers.

Hint: Does order matter?

```
def donut(d, f):  
    """  
    >>> donut(12, 1)  
    1  
    >>> donut(12, 2)  
    13  
    >>> donut(12, 12)  
    1352078  
    >>> donut(0, 0)  
    1  
    """  
    if _____:  
        return _____  
  
    if _____:  
        return _____  
  
    return _____
```